General Capabilities

Critical and creative thinking

In Technologies, students develop critical and creative thinking as they imagine, generate, iterate and critically evaluate ideas. Students analyse problems, refine concepts and reflect on the decision-making process by engaging in systems, design and computational thinking.

Students think critically and creatively while considering how data, information, systems, materials, tools and equipment (past and present) impact on our lives, and how these elements might be better designed and managed. Experimenting, drawing, modelling, designing and working with equipment and software helps students to build their visual and spatial thinking and to create solutions, products, services and environments.

Ethical understanding

Technologies develops students' capacity to understand and apply ethical and socially responsible principles when managing projects and collaborating with others and creating, sharing and using technologies. Using an ethical lens, they investigate past, current and future local, national, regional and global technological priorities. Understanding the protection of data, intellectual property and individual privacy helps students to be respectful creators.
Students explore complex issues and consider possibilities and ethical implications. They learn about safe and ethical procedures, and consider the rights of others and their responsibilities in using sustainable practices. Students learn to appreciate and value the part they play in the social and natural systems they live in.

Literacy

In Technologies, students develop literacy by interpreting a wide range of practical texts including technical manuals and specifications. They evaluate content in diverse formats, analyse data and reports and navigate complex information. Students learn to use technical symbols, icons and terminology, adapting language for both general and specialised contexts. They produce purpose-driven texts such as annotated drawings, software guides and project plans tailored to specific needs and audiences.

Numeracy

In Technologies, students develop the capacity to interpret and use mathematical knowledge and skills in a range of real-life situations. They use numbers to calculate, measure and estimate; interpret and draw conclusions from statistics; measure and record throughout the process of generating and iterating ideas; develop, refine and test concepts; and cost and sequence when making products and managing projects. In using software, materials, tools and equipment, students work with the concepts of numbers, geometry, scale, proportion, measurement and volume. They use three-dimensional models, create accurate technical drawings, work with digital models and use computational thinking in decision-making processes when designing and creating best-fit solutions.

Personal and social capability

Students develop personal and social capabilities in Technologies through engaging in project management and design activities in a collaborative environment. They show initiative in their learning, apply design thinking skills and make informed decisions independently. Working in teams, they develop cooperation, leadership, conflict resolution and resource-sharing skills. Embracing innovation and risk fosters resilience as they navigate uncertainty.
Students also cultivate social responsibility by considering the impacts of decisions on people, communities and environments. They deepen empathy and respect by understanding diverse user needs and reflect on how digital tools and environments such as social media affect their wellbeing, applying appropriate strategies to both face-to-face and digital environments.

Pre-primary

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning builds on the Early Years Learning Framework and each child's funds of knowledge. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Design and Technologies provides opportunities for children to explore familiar technologies and the need for design, while developing an understanding of the components and processes involved. They generate, record and share design ideas through discussion and drawing to create solutions, and use personal preferences to evaluate products.

In Pre-primary, children discover the uses of familiar technologies in everyday life in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. They explore common objects for their purpose and design features to develop understandings of designing solutions to solve a problem and meet personal needs. Children use technologies safely by observing teacher modelling or role-play to participate in either elementary engineering, plant/fibre production, food preparation or identify

materials used for clothing needs. They experiment with identifying problems and draw on their memory of a sequence of steps to complete a task, such as packing away play equipment or completing a puzzle.

Achievement Standard

By the end of the year:

Children explore familiar objects within the community and recognise design thinking skills. In Engineering principles and systems, children observe how objects move when force is applied. In Food and fibre production, children connect animal and plant products for everyday use as food and/or clothing. In Food specialisations, children explore food for colour, texture, flavour and shape, and use hygiene practices. In Materials and technologies specialisations, children explore and identify materials to use for creation and/or construction of a variety of products, including for personal needs, considering the materials' properties.

In the Design and Technologies contexts, children explore the purpose for designed solutions. They generate, record and share design ideas through discussion, drawing and modelling. Children safely use given technologies (tools and equipment), to create solutions and evaluate these using personal preferences.

Content Descriptions

Contexts

Engineering principles and systems

Objects can be moved when force is applied

For example:

- objects can be moved through actions including push, pull, bounce, slide, fall, spin and float
- the force of the action affects the movement
- a pulley as a way to move objects, such as a rope over a bar to lift items

WAPTDECEP1

Literacy Critical and creative thinking

Food and fibre production

Animals and plants have essential needs, including food, water, space and shelter

For example:

- various ways to provide homes and shelter for animals and plants
- awareness of processes for 'garden to plate'

WAPTDECFF1

Literacy Critical and creative thinking

Food specialisations

Foods come in different colours, textures, flavours and shapes

For example:

- explore the colour, texture, flavour and shape of food, and change these using food preparation techniques (slicing, grating, chopping)
- the effect of heat on rolled oats (to soften), addition of liquid (to cook and make easier to eat), addition of fruit (for colour, texture, flavour)

WAPTDECFS1

Literacy Critical and creative thinking

Appropriate hygiene practices are necessary for safe food handling

For example:

- use hygiene practices, such as washing and drying hands, securing hair, cleaning surfaces before and after handling food

WAPTDECFS2

Literacy Critical and creative thinking

Materials and technologies specialisations

Everyday objects are made using different materials

For example:

- furniture and play equipment is made from different materials, has different colours and textures
- stepping stones; walk barefoot over squares of different materials with different surfaces and textures
- materials for construction based on properties, such as solid blocks are more stable for a base than a cardboard packet
- take a barefoot walk on Country to discover different surfaces and textures

WAPTDECMT1

Literacy Critical and creative thinking

Technologies and society

Explore familiar technologies to meet personal needs

For example:

- EP&S: technology appliances or devices for personal use, such as a night light, small fan or remote-controlled toy
- F&FP: a range of footwear or clothes for specific activities, such as a sport, dance or family outing
- FS: a range of utensils required for food preparation
- M&TS: various closures or fasteners on a range of shoes and jackets to secure, and/or keep warm, such as a hook and loop fastener, and zipper

WAPTDETS1

Personal and social capability Ethical understanding

Design thinking skills

Project management

Share ideas to develop a solution

WAPTDEDTPM1

Personal and social capability

Investigating and defining

Explore the purpose for design

WAPTDEDTID1

Literacy Critical and creative thinking

Designing

Design solutions through discussion, drawing and/or modelling to meet a personal need

WAPTDEDTD1

Critical and creative thinking

Producing and implementing

Use available technologies and materials to safely create a solution

WAPTDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use personal preferences to evaluate the solution

WAPTDEDTE1

Literacy Critical and creative thinking

Year 1

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning experiences are informed by the Principles and Practices of the Early Years Learning Framework. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Design and Technologies provides opportunities for children to explore ways force generates movement in objects, or identify essential needs for animals and plants, or know sources of familiar food, or recognise properties of selected materials and develop design thinking skills to create solutions.

In Year 1, children have opportunities to explore and create solutions in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. They investigate the process of designing and producing products for personal needs.

Children have a natural curiosity about their physical, social and technological world and learn through play and experimentation. They observe, manipulate and explore objects and ideas, materials and technologies. Children explore technologies taking particular note of the materials, equipment, and ways to work safely to make products. They begin to develop an understanding that products have a purpose for their own personal needs and that of others. Children have opportunities to create solutions through guided learning, focusing on ideas and design for a personal need. They use available

technologies and materials to safely manage and create a preferred solution, and evaluate these solutions based on personal preferences.

Achievement Standard

By the end of the year:

Children explore ways familiar products are designed, and recall design thinking skills. In Engineering principles and systems, children use forces to generate movement in objects and explore ideas for the reactions observed. In Food and fibre production, children identify the essential needs of plants and animals used for production. In Food specialisations, children investigate various sources of food and ways to prepare food for eating. In Materials and technologies specialisations, children observe, explore and select materials to use for construction of various products based on the materials' properties.

In the Design and Technologies contexts, children explore opportunities when designing solutions. They communicate and develop design ideas through drawing, modelling and/or a sequence of steps. Children use given technologies (tools and equipment) and work safely to create products for preferred solutions. They use personal preferences to evaluate processes and designed solutions. Children work with others, guided by a sequence of steps, to create products for a personal need.

Content Descriptions

Contexts

Engineering principles and systems

Force generates movement in objects and can be affected by materials

For example:

- surfaces including carpet, concrete and grass affect the movement of objects
- different shapes move differently down a slope
- varying force is required to push or pull an object on different surfaces

WA1TDECEP1

Literacy Critical and creative thinking

Food and fibre production

Living things are used in everyday life for food and clothing

For example:

- fibres for clothing, such as from cotton, wool and bamboo
- ways seed planting needs to be considered to maximise growth, such as different growing conditions, plant spacings and growth patterns
- sustainable collection of food scraps for compost

WA1TDECFF1

Literacy Critical and creative thinking

Food specialisations

Familiar foods are sourced from various places

For example:

- various food sources and familiar foods, such as dairy cow – cheese, milk; hen – egg; wheat grains – flour, bread

WA1TDECFS1

Literacy Critical and creative thinking

Foods are prepared in a range of ways for consumption

For example:

- prepare selected foods for a green salad, such as washing and tearing leafy vegetables, removing stalks and breaking down broccoli florets

WA1TDECFS2

Literacy Critical and creative thinking

Materials and technologies specialisations

Properties of a material determine its selection for a specified purpose

For example:

- materials with protective qualities, such as for warmth, waterproofing, strength and durability chosen for winter clothing
- fabrics with a close weave and UV protection are used for swimwear and clothing worn in the sun
- versatility and durability of materials including plastic to make chairs
- use of opaque materials or fabrics to make shadow puppets

WA1TDECMT1

Literacy Critical and creative thinking

Technologies and society

People use technologies to create products for personal needs

For example:

- EP&S: electric blender to create a juice or smoothie for breakfast
- F&FP: tractor to dig holes for planting of trees/hedges to create shelter for animals
- FS: barbecue grill to cook food
- M&TS: materials to create an umbrella to protect from the rain and/or the sun

WA1TDETS1

Personal and social capability Ethical understanding

Design thinking skills

Project management

Share ideas and work with others to develop a solution

WA1TDEDTPM1

Personal and social capability

Investigating and defining

Explore ideas and design opportunities for a personal need

WA1TDEDTID1

Literacy Critical and creative thinking

Designing

Design solutions through drawing, modelling and/or a sequence of steps

WA1TDEDTD1

Critical and creative thinking

Producing and implementing

Use available technologies and materials to safely create a solution

WA1TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use personal preferences to evaluate the solution for a personal need

WA1TDEDTE1

Literacy Critical and creative thinking

Year 2

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning experiences are informed by the Principles and Practices of the Early Years Learning Framework. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Design and Technologies provides opportunities for children to explore ways people use selected technologies to create familiar products and environments to meet local needs and develop an understanding of the importance of managing design in creating products, systems or environments.

In Year 2, children are provided opportunities to create solutions in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Children explore ideas for design opportunities for a known user, such as a family member, and produce products using given equipment and technologies to safely create the designed solution. They have opportunities to create a range of solutions through guided learning and in collaboration with peers.

Children begin to develop design thinking skills by conceptualising possible solutions as a drawing, model, or sequence of steps in a process, and ways to control the use of selected technologies. They explore ways to manage and solve problems through

discussion, planning, sharing of ideas and working with others to develop designed solutions and create products, systems or environments.

Achievement Standard

By the end of the year:

Children explore and exemplify roles of people in designing solutions for familiar products and environments. In Engineering principles and systems, children use a range of forces to move objects within a system and observe reactions. In Food and fibre production, children make connections between different seasons and environments. In Food specialisations, children select food from local sources to create a food product. In Materials and technologies specialisations, children create products for a specified purpose, considering the properties of the materials available.

In the Design and Technologies contexts, children explore design solutions to meet local needs or design opportunities for a known user. They generate, communicate and explore ideas for designed solutions through discussion, drawing, modelling and/or sequenced steps. Children use given technologies and components to safely create products. They use personal preferences and the needs of known users to evaluate the designed solution and the process used.

Content Descriptions

Contexts

Engineering principles and systems

Force can move objects within a system

For example:

- large and small cogs work together to create movement
- the slope that marbles or small balls roll down affects their movement
- bends, 'S' curves and/or straights affect the movement of the object

WA2TDECEP1

Literacy Critical and creative thinking

Food and fibre production

Food and fibre are produced in different seasons and environments

For example:

- different foods and fibres available in each of the seasons
- plants grow in different seasons
- environment required for silk production, fish farming and/or greenhouse tomatoes
- system (process layout) for egg, wheat production or print on a T-shirt

WA2TDECFF1

Literacy Critical and creative thinking

Food specialisations

Staple foods from local sources are used to create a range of food products

For example:

- ways staple foods are used to prepare a dish, such as oats rolled for porridge; wheat flour for bread, pasta, noodles; tomatoes for sauce, pizza; rice for sushi, fried rice; potatoes and yams, as a basis for meals
- benefits of using local food, such as freshness and could be cheaper

WA2TDECFS1

Literacy Critical and creative thinking

Materials and technologies specialisations

Materials can be combined to produce a product for a specified purpose

For example:

- given a range of materials, select for product suitability, such as for a hat, toy boat or simple bat
- different types of materials, such as foam, bubble wrap to create seat cushions
- materials suitable to protect from various weather conditions, such as shade for animals or rain jackets for children
- a range of materials provided for loose parts play

WA2TDECMT1

Literacy Critical and creative thinking

Technologies and society

People use selected technologies to make familiar products and environments to meet local needs

For example:

- EP&S: design features and selected technologies of playground equipment for the needs of a child
- F&FP: hot house environment for growing food, such as tomatoes or cucumbers to meet needs of local people
- FS: selected technologies, such as an electric food mixer, measuring scales or cups and an oven to produce bread, cupcakes or banana bread
- M&TS: design shelving for extra storage in the classroom and identify appropriate technologies and materials, such as timber and metal for the designer to consider

WA2TDETS1

Personal and social capability Ethical understanding

Design thinking skills

Project management

Plan, share ideas and work with others to develop a solution for a known user

WA2TDEDTPM1

Personal and social capability

Investigating and defining

Explore ideas and design opportunities for a known user

WA2TDEDTID1

Literacy Critical and creative thinking

Designing

Design solutions through discussion, drawing, modelling and/or a sequence of steps

WA2TDEDTD1

Critical and creative thinking

Producing and implementing

Use given equipment and technologies to safely create a solution

WA2TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use personal preferences and the needs of the known user to evaluate the solution

WA2TDEDTE1

Literacy Critical and creative thinking

Year 3

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Design and Technologies builds on concepts previously acquired and students continue to develop understanding in design thinking skills, such as products to assist people with limited mobility, hearing or sight, and outlining procedures to achieve solutions.

In Year 3, students have opportunities to learn about technologies in society and to create solutions in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students are provided with opportunities to manage, develop ideas, design and make products for individual and/or local community needs.

Students apply design thinking skills to generate multiple ideas for their designed solutions. They learn to define problems using project management skills. Students experiment with appropriate work protocols and consider ways to improve, modify or adapt for different situations, including safety. They use given criteria to evaluate diagrams, technologies and the components used for the designed solution.

Achievement Standard

By the end of the year:

Students explore and recognise roles people in Design and Technologies occupations have in the local community. They explore ways technologies are designed for solutions and used to create products, services or environments to meet individual or local community needs. In Engineering principles and systems, students observe and recognise ways forces and the properties of materials affect the behaviour of objects. In Food and fibre production, students recognise food and fibre are produced to meet food and clothing needs. In Food specialisations, students select food to nourish the body for energy to move and support growth. In Materials and technologies specialisations, students select and safely use suitable materials, appropriate technologies and components to create a product to achieve a designed solution.

In the Design and Technologies contexts, students develop a sequence of steps to achieve a designed solution. They define features of a design brief and communicate ideas using labelled drawings and technical terms. Students select and safely use appropriate technologies and components to develop a solution. They use given criteria to evaluate diagrams, technologies and components used for the designed solution. Students follow a plan, communicate sequenced steps to manage and create a product, service or environment for an individual or local community need.

Content Descriptions

Contexts

Engineering principles and systems

Forces, and the properties of materials affect the behaviour of objects

For example:

- drop a range of balls from same height
- roll a marble into a group of marbles using different force
- resistance; roll marbles or types of balls down/along different surfaces, with same gradient, force and distance

WA3TDECEP1

Literacy Critical and creative thinking

Food and fibre production

Food and fibre produced to meet food and clothing needs

For example:

- basket weaving, finger knitting and/or hand sewing to produce items for use or to wear
- past and current cereal/grain production, collection and sorting
- wool production and uses, including waterproof, insulation and fire-resistant properties
- 'fascinating fibres' – up close with a microscope to observe different types of fibres, such as cotton, wool and nylon; similar examination of plants, such as carrot, celery and spinach leaves

WA3TDECFF1

Literacy Critical and creative thinking

Food specialisations

Foods are selected to nourish the body, for energy to move and support growth

For example:

- food models used to select food
- 'mindful eating' to nourish the body
- consumption of wholegrain products for energy to move
- rehydrate with water

WA3TDECFS1

Literacy Critical and creative thinking

Materials and technologies specialisations

Properties of materials, suitability and safe practice using given technologies to create a product to achieve a purpose

For example:

- magnifying glass to view the structure of materials and relate to purpose, such as non-woven textiles for waterproofing
- model raft construction from different materials, such as local grasses and pop sticks, and observe performance
- system of paper making for a product, considering design features, such as colour, strength, functionality and shape (for a small bowl)

WA3TDECMT1

Literacy Critical and creative thinking

Technologies and society

Role of people in design and technologies occupations in the local community

For example:

- EP&S: electrician provides a service to repair fans, ovens, lights and other electrical technologies used in the home
- F&FP: carer at an animal refuge uses a range of technologies to care for sick and injured animals
- FS: canteen manager organises the production of lunches using a range of kitchen technologies, such as utensils and cooking equipment
- M&TS: picture framer uses a range of materials and specialised technologies to create a photo frame or digital frame

WA3TDETS1

Personal and social capability Ethical understanding

Technologies are designed and used in products, services or environments to meet individual needs

For example:

- EP&S: ways product design and specific technologies can assist a person with limited mobility, hearing or sight
- F&FP: design environments for the growth of seedlings indoors
- FS: products designed using technologies to keep food cool and fresh in a lunch box
- M&TS: products designed for use by children or the elderly, such as kitchen utensils, or a bell for a bike, scooter or wheelchair

WA3TDETS2

Personal and social capability Ethical understanding

Design thinking skills

Project management

Communicate ideas and follow a plan with consideration of time management, to develop a solution

WA3TDEDTPM1

Personal and social capability

Investigating and defining

Define ideas and design opportunities for individual and/or local needs

WA3TDEDTID1

Literacy Critical and creative thinking

Designing

Design solutions through use of labelled drawings, technical terms and/or a sequence of steps

WA3TDEDTD1

Critical and creative thinking

Producing and implementing

Use appropriate technologies and components with given equipment and follow agreed protocols to produce a designed solution

WA3TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use given criteria to evaluate diagrams, technologies and the components used for the designed solution

WA3TDEDTE1

Literacy Critical and creative thinking

Year 4

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Design and Technologies builds on concepts previously acquired and students continue to develop an understanding of design thinking skills, such as properties of materials and outlining step-by-step procedures. They have opportunities to create a range of solutions, including the reuse, repurpose and recycle of materials, and select materials from regenerated sources.

In Year 4, students have opportunities to learn about technologies in society and the diverse roles for people in design and

technologies occupations and create solutions in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students consider the way products, services and environments are designed to meet community needs, including consideration of sustainable factors.

Students define solutions to meet specific needs and consider society's use of technologies that meet community requirements and implement project management protocols, appropriate technologies, components and equipment to produce designed solutions. They use agreed protocols and management roles to communicate ideas, plan and make decisions to develop solutions to achieve a purpose. Students use given criteria to evaluate design features, selection of resources, and decision-making processes.

Achievement Standard

By the end of the year:

Students recognise diverse roles people in Design and Technologies occupations have in the community. They consider ways products, services and/or environments are designed to achieve planned solutions that meet community needs, considering sustainable factors. In Engineering principles and systems, students recognise ways forces and properties of materials affect the behaviour of an object or system. In Food and fibre production, students explore ways technologies are used in a range of environments, cultures and/or time periods for food and natural fibre production. In Food specialisations, students select and prepare food, considering a range of physical properties. In Materials and technologies specialisations, students select materials and components for preferred properties, suitability and function for a range of purposes.

In the Design and Technologies contexts, students investigate and select resources suitable for a designed solution to achieve a given task. Students manage, communicate and define ideas using labelled drawings, appropriate technical terms, decision-making and/or a sequence of steps. They implement agreed protocols, appropriate technologies and components to produce designed solutions. Students use given criteria to evaluate design features, selected resources and solutions for a given task. They use management roles to plan, communicate ideas and make decisions to safely achieve designed solutions.

Content Descriptions

Contexts

Engineering principles and systems

Forces, and the properties of materials affect the behaviour of an object or system

For example:

- different gear levels on a bicycle
- surface of playground equipment
- surface of tyres, range of treads, surface area
- variation of force of compression when operating a bicycle pump
- spikes and studs on athletic shoes, such as sprinting shoes, football boots

WA4TDECEP1

Literacy Critical and creative thinking

Food and fibre production

Food and fibre produced in different time periods or cultures, including the technologies and equipment used

For example:

- ways people produce a variety of foods, such as yoghurt, cheese and flour
- ways people use a variety of fibres suitable for clothing, protection from sun and heat, and to keep cool (curtains, blinds)
- preserve food, such as drying fish, fruits, seeds for out of season consumption
- preserve fibres and textiles for longevity, such as regular cleaning, polish leather shoes, jackets and belts, and ways to reuse, recycle and repurpose

WA4TDECFF1

Literacy Critical and creative thinking

Food specialisations

Physical properties of food influence selection and preparation

For example:

- appearance of food, such as colour, size, shape, gloss
- texture – mouthfeel, savour the flavour, eat slowly, eat for enjoyment
- freshness – over-ripe banana suitable for baking, but not eating

WA4TDECFS1

Literacy Critical and creative thinking

Materials and technologies specialisations

Properties of materials and components for a range of purposes affect suitability and function in a system

For example:

- different properties of materials and selected components affect the function of a carry bag/basket, such as a cloth carry bag and a woven basket
- a decorative face mask, produced safely from a range of materials to achieve a given purpose, such as for a community celebration
- various materials used to construct musical instruments affect the function of the instrument

WA4TDECMT1

Literacy Critical and creative thinking

Technologies and society

Diverse roles for people in design and technologies occupations

For example:

- EP&S: designer of prams and strollers considers ways forces and materials affect mobility for a safe and smooth ride for a baby or toddler
- F&FP: school gardener assists students to design and make growing boxes for vegetables and develop an automatic water system to grow food for the school community
- FS: home fruit grower manages the growing cycle of plants till maturity, then preserves the fruit as pickles, sauces and jams
- M&TS: a woodworker designs utensils, chopping boards and children's toys, and develops a production plan to produce, evaluate and package as gifts or to sell the finished wood product

WA4TDETS1

Personal and social capability Ethical understanding

Products, services and/or environments are designed to meet community needs, including consideration of sustainable factors

For example:

- sustainable factors incorporated at the design stage for products, services and/or environments to meet community needs is effective for change, such as
  EP&S: ability to reuse, repurpose or recycle materials used for construction of wheels, frame and components for a pram or stroller
  F&FP: energy-efficient watering systems for a school garden
  FS: reuse, repurpose or recycle packaging like plastic bags and bottles, tins, and glass jars used for preservation of foods
  M&TS: select materials, including wood, from regenerated sources

- EP&S: ability to reuse, repurpose or recycle materials used for construction of wheels, frame and components for a pram or stroller
- F&FP: energy-efficient watering systems for a school garden
- FS: reuse, repurpose or recycle packaging like plastic bags and bottles, tins, and glass jars used for preservation of foods
- M&TS: select materials, including wood, from regenerated sources

WA4TDETS2

Personal and social capability Ethical understanding

Design thinking skills

Project management

Use agreed protocols and management roles to communicate ideas, plan and make decisions, to develop solutions

WA4TDEDTPM1

Personal and social capability

Investigating and defining

Define the features of a design brief and the requirements of a design task for a community need

WA4TDEDTID1

Literacy Critical and creative thinking

Investigate and select resources based on properties for the given task

WA4TDEDTID2

Critical and creative thinking

Designing

Design solutions through use of labelled drawings, technical terms, decision-making and/or a sequence of steps

WA4TDEDTD1

Critical and creative thinking

Producing and implementing

Use appropriate technologies, components and/or equipment and follow agreed protocols to produce a designed solution

WA4TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use given criteria to evaluate design features, selected resources, decision-making processes and the designed solution

WA4TDEDTE1

Literacy Critical and creative thinking

Year 5

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Design and Technologies builds on concepts previously acquired and students continue to develop an understanding of design thinking skills through collaborative planning and decision-making, and interaction with knowledgeable others. Students demonstrate an increased responsibility for project management with the implementation of agreed protocols to communicate and organise activities, individually, and in groups of varying sizes.

In Year 5, students have opportunities to learn about technologies in society, ways people in design and technologies occupations consider competing factors in the design of products, through different technologies, as they create solutions in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students are provided with opportunities to produce products, services or environments, and develop an understanding of designing solutions, considering features, such as the repurposing of components, sustainable factors and regenerative practices.

Students develop strategies to manage and communicate information and ideas using agreed ethical protocols and consider the safety aspects of working with others. They break down design briefs to define the purpose and requirements for a given task and designed solutions, considering competing factors with annotated diagrams and a sequence of steps, using technical terms and an iterative process. Students use management roles to communicate decisions, plan and manage time to develop a designed solution and reference the given criteria to evaluate.

Achievement Standard

By the end of the year:

Students identify ways people in Design and Technologies occupations consider competing factors in the design of products, services and environments. In Engineering principles and systems, students distinguish various ways forces control movement, sound or light in a product or system. In Food and fibre production, students identify ways to improve efficiency of production systems, considering sustainable factors for food and natural fibre products. In Food specialisations, students implement food preparation systems, considering ways food safety affects the selection of food for designed solutions. In Materials and technologies specialisations, students consider the properties for a range of materials, suitable components and use of appropriate technologies to achieve a designed solution for an identified purpose.

In the Design and Technologies contexts, students break down a design brief to define the purpose for a given task, including required resources, technologies and components. They manage and communicate alternative ideas using annotated diagrams and storyboards applying appropriate technical terms for designed solutions. Students use management roles to plan, manage time and communicate decisions, and implement agreed protocols when using technologies and components to produce a designed solution. They use given criteria to evaluate design features, competing factors and the sequence of steps used.

Content Descriptions

Contexts

Engineering principles and systems

Forces can control motion, sound or light in a product or system

For example:

- movement of a scooter, skateboard or bike through human force
- sound projected using a cone shaped object or musical instrument
- intensity of light or sound is controlled by variable forces, such as with a dimmer switch or volume control
- movement of air by a hair dryer, air conditioner or fan

WA5TDECEP1

Literacy Critical and creative thinking

Food and fibre production

Food and fibre production in environments for sustainable and regenerative practices

For example:

- circular recycling of clothing, such as uniforms
- the relationship between plant types and their environment, such as for hydroponics
- bamboo production for food, fabrics, furniture and tools
- sustainable animal and plant fibre production systems, such as forestry management in timber plantations

WA5TDECFF1

Literacy Critical and creative thinking

Food specialisations

Systems for food preparation and food safety affect selection of food for meals/products

For example:

- 'assembly line' system of prepared ingredients for lunch foods, such as sandwiches and wraps
- a food safety system could include
  clean surfaces, utensils, hands
  separate raw and cooked foods
  storage at appropriate temperatures, in containers
  food suitable for a lunch box or family picnic to include babies, toddlers, adults and/or seniors

- clean surfaces, utensils, hands
- separate raw and cooked foods
- storage at appropriate temperatures, in containers
- food suitable for a lunch box or family picnic to include babies, toddlers, adults and/or seniors

WA5TDECFS1

Literacy Critical and creative thinking

Materials and technologies specialisations

Properties for a range of materials, related components and use of given technologies to achieve a purpose

For example:

- various fibres combined to twist or plait a rope/belt, considering function, durability and aesthetics for fabric production
- properties of material used to protect and transport items, such as a laptop, cricket bat or a dozen eggs
- range of materials, like paper or board with plain or patterned surfaces used to create (origami) shapes, designed to achieve a purpose, such as kite making, gift cards and decorations

WA5TDECMT1

Literacy Critical and creative thinking

Technologies and society

People in design and technologies occupations consider competing factors in the design of products, services and/or environments

For example:

- EP&S: engineer considers competing factors to design for sustainability, including ways to repurpose resources and components used in products, such as electrical appliances and solar panels
- F&FP: gardener or horticulturalist considers competing factors to repurpose vacant land for a community need, such as a sensory garden; grows seedlings for food and a peaceful environment
- FS: chef and wait staff consider competing factors to repurpose a classroom for a community lunch
- M&TS: fashion designer considers competing factors to repurpose uniforms, garments, and accessories for identified community needs

WA5TDETS1

Personal and social capability Ethical understanding

Technologies are used in the design of products, and implementation of services and environments

For example:

- EP&S: the combinations of technologies required for a transport system, such as trains, buses and trucks to deliver services for the community
- F&FP: the combination of technologies and resources required to design a system to grow food for a community garden
- FS: the combination of technologies and resources required to design a system to deliver food produce to the school canteen
- M&TS: the combination of materials and technologies are used to produce different versions of popular products to be exclusive, unique, fashionable, distinctive and/or for safety

WA5TDETS2

Personal and social capability Ethical understanding

Design thinking skills

Project management

Use agreed protocols and management roles to communicate decisions, plan and manage time, to develop designed solutions

WA5TDEDTPM1

Personal and social capability

Investigating and defining

Break down a design brief to define the purpose and requirements for a given task

WA5TDEDTID1

Literacy Critical and creative thinking

Investigate and select resources based on properties and functions for the given task

WA5TDEDTID2

Critical and creative thinking

Designing

Design solutions considering competing factors, with annotated diagrams, storyboards and/or a sequence of steps, using technical terms and an iterative process

WA5TDEDTD1

Critical and creative thinking

Producing and implementing

Use technologies, components and/or equipment to implement agreed protocols to produce a designed solution

WA5TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use given criteria to evaluate design features, with consideration of competing factors, processes and the designed solution

WA5TDEDTE1

Literacy Critical and creative thinking

Year 6

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Design and Technologies builds on concepts previously acquired and students continue to develop an understanding of design thinking skills. Students experiment with a variety of materials to investigate the advantages of different representational forms for different purposes and situations, such as for engineering systems, food and fibre production, food preparation systems, and suitability and functionality in a product.

In Year 6, students have opportunities to learn about technologies in society and ways people in design and technologies occupations address competing considerations, including sustainable factors, as they create solutions in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students consider ways competing technologies are used in the production of products and develop an understanding of designs for services and environments for community needs.

Students understand and appreciate different points of view, develop the ability to think in more abstract terms, and undertake sustained activities for longer periods. They work collaboratively and communicate decisions to develop agreed protocols, set goals, manage competing factors, resources and time to develop solutions for a given task. Students design alternative solutions, achieved through an iterative process. They develop negotiated criteria to evaluate design features, selected technologies, functionality and consideration of constraints of the designed solution to achieve a purpose.

Achievement Standard

By the end of the year:

Students identify ways people address competing considerations, including sustainable factors in the design of products, services and environments. In Engineering principles and systems, students connect ways electrical energy and forces can control motion, sound or light in a product or system. In Food and fibre production, students consider design features, consumer demand and managed environments for food and natural fibre systems. In Food specialisations, students consider food choices, consumer demands, and preparation systems in the design of a meal/food product. In Materials and technologies specialisations, students consider the properties of selected materials, technologies and production systems, to plan and develop a designed solution for an identified purpose.

In the Design and Technologies contexts, students define ways competing considerations affect decisions on the selection of technologies, resources, techniques and sustainable factors in the design of a solution. Students manage and consider alternative solutions through discussion, critical thinking, modifying processes and initial design ideas using a range of appropriate technical terms, technologies and techniques. They implement production plans and safe procedures when using a variety of technologies and components to produce solutions. Students develop negotiated criteria to evaluate design features, selected technologies, processes and functionality of the designed solution. They use agreed protocols to set goals, manage competing factors, resources and time, and communicate decisions and solutions for a given task.

Content Descriptions

Contexts

Engineering principles and systems

Forces and electrical energy can control motion, sound or light in a product and/or system

For example:

- an electrical circuit, including switches or movement sensors for output to light-emitting diodes (LEDs), and buzzers for sound
- addition of light, using circuits, to decorate clothing, costumes, hats and shoes
- catapults work to transfer force, as do cams, camshafts, cranks and worm drives

WA6TDECEP1

Literacy Critical and creative thinking

Food and fibre production

Food and fibre production systems for products, considering design features, consumer demand and managed environments

For example:

- variables within a system, such as quantity and quality of fertilisers in separate plots/pots for optimum plant growth
- twist or spin together one or more fibres to create yarns, to produce a fabric
- compare hothouse environments, trellis espalier, and open field growth of tomatoes
- animal welfare considerations in various environments
- soil health/topsoil degradation affected by drought and floods
- past, current and future needs are considered through cotton fibre production and processes

WA6TDECFF1

Literacy Critical and creative thinking

Food specialisations

Food choices, consumer demands and preparation systems affect the use of a food in a meal/product

For example:

- food selection guides, such as the Australian Guide to Healthy Eating to make food choices
- 'what to eat' and 'how we eat'; mindful eating; regular meals versus grazing
- ethical considerations affect food choices for inexpensive and inventive recipes to provide food that is nutritious and nourishing

WA6TDECFS1

Literacy Critical and creative thinking

Materials and technologies specialisations

Properties of selected materials, technologies, and production systems affect suitability and functionality in a product

For example:

- combination of materials to secure and identify a bike or personal schoolbag/backpack
- combination of materials and technologies to produce a photo frame using a planned production system
- combination of materials and components suitable for babies, toddlers and children

WA6TDECMT1

Literacy Critical and creative thinking

Technologies and society

People in design and technologies occupations address competing considerations, including sustainable factors in the design of products, services and environments

For example:

- EP&S: engineer considers competing factors, such as design features, alternative technologies and a range of components suitable to develop or adapt an environment, like the assembly area for the school community, including improved access, lighting, seating and overall comfort
- F&FP: food producer or clothing manufacturer considers competing factors, such as cost, access to preferred resources, water constraints, transport time and secure storage in the design of the selected product
- FS: chef considers competing factors, such as access to seasonal produce including sensory properties, consumer values, cost, reliable and safe transport, storage and staff skills in the design of the café menu
- M&TS: local government services consider sustainable features of selected materials, and competing factors, such as cost, suitable access and safety in the design of parks, gardens and playgrounds for all community members

WA6TDETS1

Personal and social capability Ethical understanding

Competing technologies are used for the design of products, services and environments for community needs

For example:

- EP&S: engineer considers ways competing technologies are used in public transport, media or communication systems within the local and wider community
- F&FP: a food or fibre producer considers competing technologies in the design of gardens or production systems to increase production, such as systems to weed, fertilise, water, harvest and store
- FS: a takeaway food producer considers competing technologies in production and packaging of food ready for sale considering food waste reduction and sustainable features of the packaging used
- M&TS: a local boilermaker who fabricates sheet metal into large containers considers competing technologies including safety, assembly, installation, transport and construction components in the design of a boiler system for a remote mine site

WA6TDETS2

Personal and social capability Ethical understanding

Design thinking skills

Project management

Use agreed protocols to set goals, manage competing factors, resources and time, to plan, develop and communicate decisions, when developing designed solutions

WA6TDEDTPM1

Personal and social capability

Investigating and defining

Break down a design brief to define the purpose, requirements and constraints for a given task

WA6TDEDTID1

Literacy Critical and creative thinking

Investigate and select resources considering constraints, properties and functions appropriate for the given task

WA6TDEDTID2

Critical and creative thinking

Designing

Design alternative solutions achieved through an iterative process, including critical thinking, graphical representations, use of a range of technologies, techniques, technical terms and/or a sequence of steps

WA6TDEDTD1

Critical and creative thinking

Producing and implementing

Use a range of technologies, components and/or equipment to implement agreed protocols to produce a designed solution

WA6TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Develop negotiated criteria to evaluate design features, graphics, selected technologies, processes and functionality with consideration of constraints for the designed solution

WA6TDEDTE1

Literacy Critical and creative thinking

Year 7

Year Level Description

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Learning and teaching programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Design and Technologies focuses on further development of understanding and design thinking skills in ways products evolve locally to achieve designed solutions. Students begin to develop an interest in particular fields of knowledge, such as engineering, food and fibre production, food systems and various materials and their use.

In Year 7, students have opportunities to learn about technologies in society and ways people in design and technologies occupations consider competing factors, social and ethical influences and existing technologies in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students are provided with opportunities to investigate ways products, services and environments evolve locally with the use of various technologies.

Students are provided with opportunities to manage and create a range of designed solutions to achieve a specific purpose. They extend understanding of the vital role and ways design and technologies are incorporated in everyday life. When defining problems, students identify the key elements of the problem, the intended purpose and ways competing factors and constraints are at play. They manage and design increasingly complex processes and solutions with given technologies and techniques, considering social and ethical influences. Students collaborate and implement agreed protocols when using a range of technologies, components and equipment to produce design solutions. They plan and manage individual and team projects with autonomy. Students consider ways of managing the exchange of ideas, time and available resources, and use given contextual criteria to evaluate design processes and solutions.

Achievement Standard

By the end of the year:

Students consider ways products, services and/or environments evolve locally. They recognise competing factors, including social and ethical influences and existing technologies for designed solutions. In Engineering principles and systems, students use motion, force and energy to manipulate and control engineered systems. In Food and fibre production, students identify features of production systems including managed environments to produce local products and achieve designed solutions. In Food specialisations, students recognise ways nutritional values and sensory properties of food determine preparation techniques and presentation of a designed solution. In Materials and technologies specialisations, students identify ways the properties of materials, specialised technologies and production processes influence designed solutions to achieve quality and safely produced products.
In the Design and Technologies contexts, students investigate and define the purpose for a given task and design solutions by considering constraints, social and ethical factors, available specialised technologies and ways products evolve locally. Students implement agreed protocols using a range of techniques, technologies, components and processes to produce designed solutions. They apply given contextual criteria to evaluate design processes and solutions. Students plan, develop and communicate ideas using project management processes/skills, considering time and available resources to achieve solutions.

Content Descriptions

Contexts

Engineering principles and systems

Force, motion and energy, including light and/or sound and/or heat and/or wind are used to control engineered systems

Social and ethical considerations for the design and development of engineered products and systems, including ways products evolve locally to achieve designed solutions

For example:

- control of loads with lifting devices using simple or compound pulleys or gearing
- elements involved in changing speed of a vehicle over a distance
- energy audit of family home, considering time of day, appliances in use to determine energy consumption and ways to conserve energy
- considerations, such as
social – develop engineered products to assist in everyday activities, such as ways to open canned food, put shoes and socks on to assist the infirm, family and community members, and consideration of regulatory responsibilities
ethical – use engineered systems to develop ideas to assist others considering sustainable factors, use of limited resources and cultural values

- social – develop engineered products to assist in everyday activities, such as ways to open canned food, put shoes and socks on to assist the infirm, family and community members, and consideration of regulatory responsibilities
- ethical – use engineered systems to develop ideas to assist others considering sustainable factors, use of limited resources and cultural values

WA7TDECEP1

Literacy Critical and creative thinking Personal and social capability Ethical understanding

Food and fibre production

Features of production systems, including managed environments, regulatory requirements for quality and safely produced food and/or fibre products

Social and ethical considerations for the design and development of food products and/or fibre products or a combination, including ways products evolve locally to achieve designed solutions

For example:

- milk production from farm to fridge
- cotton/linen production from seed to fabric
- wool production processes compared with synthetic fibre production
- considerations, such as
  social - shared food, eating with others, mindful eating, eating for nourishment of the body
  ethical
  food from different cultures; communities and celebrations
  managed environments to produce plant fibres, such as cotton production; and food, such as egg and meat production

- social - shared food, eating with others, mindful eating, eating for nourishment of the body
- ethical
  food from different cultures; communities and celebrations
  managed environments to produce plant fibres, such as cotton production; and food, such as egg and meat production

- food from different cultures; communities and celebrations
- managed environments to produce plant fibres, such as cotton production; and food, such as egg and meat production

WA7TDECFF1

Critical and creative thinking Personal and social capability Ethical understanding

Food specialisations

Sensory properties and nutritional value of foods determine preparation, production and presentation techniques

Social and ethical considerations for the design and development of meals and specialised food products, including ways products evolve locally to achieve designed solutions

For example:

- familiar food sources with macro nutrients
- menu plans based on the Australian dietary guidelines
- considerations, such as
  social – shared food, eating with others, mindful eating, eating for nourishment of the body; consider the roles and occupations of people who produce, prepare and provide food
  ethical – food in society; food waste; food security, that is food for all people, have physical and economic access to sufficient safe and nutritious food

- social – shared food, eating with others, mindful eating, eating for nourishment of the body; consider the roles and occupations of people who produce, prepare and provide food
- ethical – food in society; food waste; food security, that is food for all people, have physical and economic access to sufficient safe and nutritious food
- preference for locally produced food products, such as the Buy West Eat Best promotional campaign
- 'panic buying' in times of events, such as floods, war or pandemic may cause shortage of food supplies

WA7TDECFS1

Literacy Critical and creative thinking Personal and social capability Ethical understanding

Materials and technologies specialisations

Properties of combined materials, features of production systems, given components, tools and equipment for quality, safely produced products

Social and ethical considerations for the design and development of products using specialised technologies, including ways products evolve locally to achieve designed solutions

For example:

- embellishment and decorative process, using technologies, such as an electronic cutting machine, laser cutting for street numbers, sewing machine for indoor or outdoor cushions, lathe for wooden serving tray or bowl
- tote bag or draw string bag for a specific purpose, such as carrying swimwear, athletics uniform or equipment, food and drink
- embossed name tag for a pet collar, produced using either metal, textile or wood material and technologies (specific tools and equipment) for embossing
- label or decorate items, such as a metal plate using an engraving tool, a tote bag using an assortment of fibres and free-form machining, a wooden box using a wood burning tool or other alternative, suitable technologies
- considerations, such as

social factors – objects, items or clothing produced for individuals and/or community celebrations
ethical factors – cultural respect for choice of embellishment designs; incorporate sustainable factors at the design phase, such as design to assemble, disassemble and reuse components

• social factors – objects, items or clothing produced for individuals and/or community celebrations
• ethical factors – cultural respect for choice of embellishment designs; incorporate sustainable factors at the design phase, such as design to assemble, disassemble and reuse components

WA7TDECMT1

Literacy Critical and creative thinking Personal and social capability Ethical understanding

Technologies and society

People in design and technologies occupations consider competing factors, social and ethical influences and existing technologies for designed solutions

For example:

• choose an inventor/designer (local, national, international, past/contemporary), such as Leonardo Da Vinci, Thomas Edison or Prue Acton and review the design techniques and technologies used to develop and invent products and systems
• decision-making processes to select existing technologies, considering social and ethical influences
• consider sustainable elements and competing factors at the designing phase of creating a solution

WA7TDETS1

Personal and social capability Ethical understanding

Products, services and/or environments evolve locally through the application of technologies

For example:

• timeline to show the evolution of a technology and/or product, such as telescopes, agricultural machinery, processed foods (breakfast cereal), sewing machines or woodworking equipment
• value of local knowledge considering culture, skills and resources in the evolution of technologies

WA7TDETS2

Personal and social capability Ethical understanding

Design thinking skills

Project management

Plan, develop and communicate, using project management processes, considering time and available resources to achieve solutions

WA7TDEDTPM1

Critical and creative thinking Personal and social capability

Investigating and defining

Investigate and define the problem and requirements of a given design brief

WA7TDEDTID1

Literacy Critical and creative thinking

Break down a given design brief, identifying and defining the purpose and competing considerations

WA7TDEDTID2

Critical and creative thinking

Consider given technologies, resources and/or components to develop solutions

WA7TDEDTID3

Critical and creative thinking

Designing

Design processes and solutions with given technologies and techniques, using appropriate technical terms

WA7TDEDTD1

Producing and implementing

Implement agreed protocols and use a range of technologies, components and/or equipment to produce designed solutions

WA7TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use given contextual criteria to evaluate design processes and solutions

Year 8

Year Level Description

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Learning and teaching programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Design and Technologies focuses on enhancing the understanding of design thinking skills to achieve solutions, including the use of a wide range of technologies, materials and systems to broaden experiences and involvement in local and regional projects.

In Year 8, students have opportunities to learn about technologies in society and ways people design for change, considering ethical and sustainable factors, available technologies and systems for designed solutions in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students consider ways products, services and/or environments are designed and developed with creative and innovative application of technologies.

Students have opportunities to investigate a problem for a given need or opportunity, considering ethical and sustainable factors for the design and development of products and systems, including economic factors, use of locally or regionally sourced materials based on reliable supply chains to achieve designed solutions. They establish project management procedures that minimise risk and consider safety and efficiency, using materials, components and systems in combination with specialised technologies for the design, development and production of products, systems and/or environments. Students collaborate to plan, develop ideas and communicate, using project management processes to consider time, resources and costs, and the incorporation of student developed contextual criteria to assess design processes to achieve the desired solution.

Achievement Standard

By the end of the year:

Students recognise ways products, services and/or environments are designed and developed with creative and innovative use of technologies. They consider ethical and sustainable factors to design solutions for a local or regional need, or opportunity. Designed solutions consider economic factors, locally or regionally sourced materials and reliable supply chains. In Engineering principles and systems, students use force, motion and energy to control and manipulate engineered systems, and produce products. In Food and fibre production, students consider ways competing factors influence the design of food and fibre production systems. In Food specialisations, students recognise ways nutritious, sustainable diets and the physical properties of food determine processing techniques for designed solutions. In Materials and technologies specialisations, students recognise how the selection of materials, components, systems and specialised technologies influence processes to achieve designed solutions for an identified need.

In the Design and Technologies contexts, students investigate and develop a design brief for a given need or opportunity to achieve a specific purpose. They consider a range of technologies, techniques, resources, use appropriate technical terms and communicate alternative ideas to develop designed solutions. Students implement agreed protocols, using a range of components and processes to produce designed solutions. They develop contextual criteria to evaluate design processes and solutions. Students plan, develop and communicate using project management processes/skills to achieve solutions.

Content Descriptions

Contexts

Engineering principles and systems

Force, motion and energy are used to control and manipulate engineered systems

Ethical and sustainable considerations for the design and development of engineered products and systems, including economic factors, use of locally or regionally sourced materials and reliable supply chains to achieve designed solutions

For example:

- manipulation of forces, energy and motion for the remodelling of a local skate park
- manipulation of wind force on a model wind power generator affects energy generation
- balanced and unbalanced forces for motion, such as spring balance, and amusement park rides, including the roller coaster
- control sensing for light, heat or mechanical signals results in varied output/s in systems
- sustainable sources of force and energy; management of excess production of energy, including storage
- sequence of steps (energy supply chains) for utilising locally sourced energy for electricity production
- considerations, such as

ethical

system for sharing power or water in the community during disruptions and shortage

sustainable
individual responsibility, such as conserving energy or water in the household
collective endeavours, such as local community plans to generate and conserve energy or water

• ethical
system for sharing power or water in the community during disruptions and shortage

• system for sharing power or water in the community during disruptions and shortage
• sustainable
individual responsibility, such as conserving energy or water in the household
collective endeavours, such as local community plans to generate and conserve energy or water

• individual responsibility, such as conserving energy or water in the household
• collective endeavours, such as local community plans to generate and conserve energy or water

WA8TDECEP1

Literacy | Critical and creative thinking | Personal and social capability | Ethical understanding

Food and fibre production

Process for selection of foods and fibres, components, and systems, including for managed environments, to produce products

Ethical and sustainable considerations for the design and development of food products and/or fibre products or a combination, including economic factors, use of locally or regionally sourced food and fibre products, and reliable supply chains to achieve designed solutions

For example:

• sustainable production processes
source local produce and resources
management of waste materials

• source local produce and resources
• management of waste materials
• design and process selected food and fibre for a specified purpose
• considerations, such as
ethical – production of food and fibre in managed environments; demand for land use, such as primary production, forest regeneration, housing
sustainable
individual responsibility; be involved in or develop ways to use resources efficiently, such as monitor water systems for optimum plant and animal growth
collective endeavours; community insistence for circular and regenerative, sustainable fibre-based products; community gardens for food and fibre products

• ethical – production of food and fibre in managed environments; demand for land use, such as primary production, forest regeneration, housing
• sustainable
individual responsibility; be involved in or develop ways to use resources efficiently, such as monitor water systems for optimum plant and animal growth
collective endeavours; community insistence for circular and regenerative, sustainable fibre-based products; community gardens for food and fibre products

• individual responsibility; be involved in or develop ways to use resources efficiently, such as monitor water systems for optimum plant and animal growth
• collective endeavours; community insistence for circular and regenerative, sustainable fibre-based products; community gardens for food and fibre products

WA8TDECFF1

Critical and creative thinking | Personal and social capability | Ethical understanding

Food specialisations

Nutritious and sustainable diets, and physical properties of food determine processing techniques

Ethical and sustainable considerations for the design and development of specialised food products and systems, including economic factors, locally or regionally sourced produce and reliable supply chains to achieve designed solutions

For example:

• nutritious diets – influences on the choices consumers make about food and impact on their wellbeing
• sustainable diets – management of food waste, such as using leftover food, and sourcing local, fresh, seasonal food
• food supply chains and processing systems for a food produced locally or regionally
• considerations, such as
ethical – food in society, preferred food preparation techniques; where food is sourced and processed; inexpensive and inventive recipes for food that is nutritious and nourishing
sustainable
individual responsibility; ways to use leftover foods, implement appropriate storage and serving sizes to reduce food waste; packaging designs for disassembly and reuse
collective endeavours; community systems to reduce food waste, recycle packaging materials; role of community gardens

• ethical – food in society, preferred food preparation techniques; where food is sourced and processed; inexpensive and inventive recipes for food that is nutritious and nourishing
• sustainable
individual responsibility; ways to use leftover foods, implement appropriate storage and serving sizes to reduce food waste; packaging designs for disassembly and reuse
collective endeavours; community systems to reduce food waste, recycle packaging materials; role of community gardens

• individual responsibility; ways to use leftover foods, implement appropriate storage and serving sizes to reduce food waste; packaging designs for disassembly and reuse
• collective endeavours; community systems to reduce food waste, recycle packaging materials; role of community gardens

WA8TDECFS1

Literacy | Critical and creative thinking | Personal and social capability | Ethical understanding

Materials and technologies specialisations

Materials, components and systems, in combination with specialised technologies for the design, development and safe production of products

Ethical and sustainable considerations for the design and development of specialised products and systems, including economic factors, locally or regionally sourced materials and reliable supply chains to achieve designed solutions

For example:

• the use of a range of specialised technologies, such as tools and equipment for the production of a selected design solution
• sustainable production processes, such as management of waste materials, including a 'measure twice, cut once' approach
• land yacht – incorporate different materials for different purposes, such as metal frame (strength),
• fabric sails (flexibility), and wooden wheels (decorative)
• photo/picture frames – metal, timber, acrylic or combination of materials and technologies
• items to achieve specific purposes
lighting, such as a safe, lightweight candle holder/lantern
leisure wear and comfort, such as drawstring shorts, T-shirt
puzzles for children, such as decorated wooden jigsaw puzzles
considerations, such as
ethical factors – use of materials in society
sustainable factors – individual responsibility; collective endeavours

• lighting, such as a safe, lightweight candle holder/lantern
• leisure wear and comfort, such as drawstring shorts, T-shirt
• puzzles for children, such as decorated wooden jigsaw puzzles
• considerations, such as
ethical factors – use of materials in society
sustainable factors – individual responsibility; collective endeavours

• ethical factors – use of materials in society
• sustainable factors – individual responsibility; collective endeavours

WA8TDECMT1

Critical and creative thinking | Personal and social capability | Ethical understanding

Technologies and society

People design for change considering ethical and sustainable factors, available technologies and systems for designed solutions, locally and regionally

For example:

• ways sustainable factors can be incorporated at the designing phase, including preferred technologies, and select components for reuse, longevity, circular economy, risk assessment and impact on known users

- ways ethical issues and social cohesion supports positive change in society and is achieved through working together, allocating roles in the development and production phases, sharing common goals and responsibilities

WA8TDETS1

Personal and social capability Ethical understanding

Products, services and/or environments are designed and developed with creative and innovative application of technologies

For example:

- an existing invention for an individual, community or organisation has evolved over time, reflecting on changed technologies, such as the advancement of lighting from filament to LED, from harvesting crops with hand tools to mechanisation, increased 'seasonal' produce availability with the evolution of processing and transport options, progress of engines and tools with the evolution of sources of energy

WA8TDETS2

Personal and social capability Ethical understanding

Design thinking skills

Project management

Plan, develop and communicate, using project management processes, considering time, resources and costs to achieve solutions

WA8TDEDTPM1

Literacy Critical and creative thinking

Investigating and defining

Investigate a problem for a given need or opportunity

WA8TDEDTID1

Literacy Critical and creative thinking

Develop a design brief for a given need or opportunity

WA8TDEDTID2

Critical and creative thinking

Consider technologies, resources and/or components to develop solutions, identifying constraints

WA8TDEDTID3

Critical and creative thinking

Designing

Design processes and solutions considering a range of technologies and techniques, using appropriate technical terms

WA8TDEDTD1

Critical and creative thinking

Producing and implementing

Implement agreed protocols, a range of technologies, techniques, components and processes to produce designed solutions

WA8TDEDTPI1

Critical and creative thinking Personal and social capability

Evaluating

Use student-developed contextual criteria to evaluate design processes and solutions

WA8TDEDTE1

Literacy Critical and creative thinking

Year 9

Year Level Description

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Design and Technologies enhances development of the understanding and application of design thinking skills precisely and accurately to describe problems, and the use of functional properties to develop solutions. It also focuses on engaging students with specialised learning, considering enterprising behaviours and entrepreneurial activities.

In Year 9, students have opportunities to learn about technologies in society and ways people consider social, ethical and sustainable factors, and use of specialised technologies in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students explore ways products, services and environments are designed and developed, considering economic factors and alternative technologies to achieve designed solutions for a specified community need.

Students investigate and analyse problems to define a range of technologies, resources and components required to develop ideas and solutions to design alternative solutions. They consider available technologies, usability, aesthetics and application of appropriate technical terms. Students develop a strong awareness of social, ethical and sustainable considerations for the design and development of engineered products, specialised food and fibre products, food production systems or materials-based products and systems, including consumer values and management of resources to achieve designed solutions for a specified community need. They manage projects, using suitable technologies, with an agile and collaborative approach and use management processes, considering time, risk, economic and sustainable factors. Students evaluate design processes and solutions against student developed criteria, including social, ethical and sustainable considerations.

Achievement Standard

By the end of the year:

Students consider ways social, ethical and sustainable factors affect the development of designed solutions for products, services and environments to meet community needs. In Engineering principles and systems, students consider properties of materials and the influencing factors of force, motion and energy for a designed solution. In Food and fibre production, students consider ways competing factors, including social, environmental and economic, influence design features and function of specialised food and fibre for designed solutions. In Food specialisations, students consider ways nutrition, sensory properties, global tastes, packaging and labelling responsibilities influence development of specialised products to achieve designed solutions. In Materials and technologies specialisations, students identify ways the properties of materials, components, systems and specialised technologies are used to develop designed solutions.

In the Design and Technologies contexts, students ideate a problem and define the needs of an end user to develop a design brief for a solution. They investigate a range of technologies, resources and components required to develop ideas and solutions, with consideration of constraints. Students select, implement and test a range of technologies, techniques and processes to produce designed solutions and/or prototypes. They evaluate design processes and solutions against student-developed criteria including social and ethical factors. Students manage projects, using suitable technologies, an iterative and collaborative approach, and consider time, risk, economic and sustainable factors.

Content Descriptions

Contexts

Engineering principles and systems

Properties of materials, combined with force, motion and energy influence the design of engineered products and systems

Social, ethical and sustainable considerations for the design and development of engineered products and systems, including consumer and/or producer values and management of resources to achieve designed solutions for a specified community need

For example:

- forces involved in bridge and tunnel construction for road and rail infrastructure
- management of engineered systems to control combined forces and materials in the construction of train haulage systems and multistorey buildings
- engineered road systems to construct safe and efficient traffic flow, including fuel, time and resource efficiencies
- social responsibility in the design of engineered systems for specific needs, such as old age, to develop products like prostheses, mobility aids, and modified kitchen utensils to aid grip, movement, hearing and sight
- autonomous guided vehicles (AGVs)
- radio-controlled limited range drone

WA9TDECEP1

Literacy Critical and creative thinking Personal and social capability Ethical understanding

Food and fibre production

Competing factors influence the design and function of specialised food and fibre products and systems

Social, environmental, economic and sustainable considerations for the design and development of specialised food and/or fibre products and systems, including consumer and/or producer values and management of resources to achieve designed solutions for a specified community need

For example:

- social
  clothing, textile and food regulatory responsibilities to protect consumers, such as pasteurisation of milk, safe colour dyes for fabric production to create garments
  change in consumer and/or producer values, such as varieties of milk, natural fibre preferences, production of eggs, milk and meat in managed environments

- clothing, textile and food regulatory responsibilities to protect consumers, such as pasteurisation of milk, safe colour dyes

for fabric production to create garments
• change in consumer and/or producer values, such as varieties of milk, natural fibre preferences, production of eggs, milk and meat in managed environments
• environmental
nature-positive systems to support biodiversity, such as heirloom
varieties of plants; forestry production
carbon zero food and/or fibre production
role of fertiliser to increase yield

• nature-positive systems to support biodiversity, such as heirloom
• varieties of plants; forestry production
• carbon zero food and/or fibre production
• role of fertiliser to increase yield
• economic
source of raw materials and impact on cost of production and final product in the market
food security at all times, for all people
economic access to sufficient nutritious food to meet dietary needs, at all times
sourcing of adequate safe and nutritious food

• source of raw materials and impact on cost of production and final product in the market
• food security at all times, for all people
economic access to sufficient nutritious food to meet dietary needs, at all times
sourcing of adequate safe and nutritious food

• economic access to sufficient nutritious food to meet dietary needs, at all times
• sourcing of adequate safe and nutritious food

WA9TDECFF1

Critical and creative thinking Personal and social capability Ethical understanding

Food specialisations

Wet and dry processing techniques and effect on nutrition, considering demographic groups, food safety including regulatory responsibilities for packaging and labelling; storage and transport of food; food enhanced for nutrition and sensory properties, global tastes and perceptions

Social, ethical and sustainable considerations for the design and development of specialised food products and systems, including consumer and/or producer values and management of resources to achieve designed solutions for a specified community need

For example:

• food enhanced for flavour, appearance, nutritive value, and impact on allergies/intolerances
• ways processing techniques affect the nutritive value of a meal, such as protein content for older people
• food safety could be compromised if food substitution occurs during times of shortage, including food fraud (deliberate and accidental) and impact on food allergies and intolerances
• food security – influence of ethical considerations to integrate secure, equitable access to healthy, safe food and sustainable diets at all times, for all people and communities
• circular sustainable supply chains, or 'close the loop', to recycle more and waste less, particularly for food solutions and packaging

WA9TDECFS1

Critical and creative thinking Personal and social capability Ethical understanding

Materials and technologies specialisations

Properties of materials, components, specialised tools, equipment and technologies used in the design and development of production systems to produce materials-based products

Social, ethical and sustainable considerations for the design and development of specialised materials-based products and systems, including consumer and/or producer values and management of resources to achieve designed solutions for a specified community need

For example:

• features required for accurate product labelling and instructions for product use, storage, safety and optimum performance
• accurate list management of materials and components used to produce an item of jewellery, a soft toy, wooden box or scooter, from recycled items
• technologies used to develop items for specified markets, school, community, artisan events and/or displays, such as gardening tools, hanging baskets (weaving, macramé), pot stands, racks, boxes, jewellery, baby/toddler clothes

WA9TDECMT1

Literacy Critical and creative thinking Personal and social capability Ethical understanding

Technologies and society

People consider social, ethical and sustainable factors, and use specialised technologies for designed solutions to address community needs

For example:

- explore and design alternative sustainable methods of transport and storage, source replacement resources, innovate production systems and adopt a creative approach to meet community needs
- obsolescence within products, services and systems impacts society, materials, sustainable factors and the environment
- factors to reduce, recycle, reuse, repurpose and remove obsolescence in designing sustainable products, services or environments for the community
- apply conflict resolution skills and work collaboratively to achieve a common goal

WA9TDETS1

Personal and social capability Ethical understanding

Products, services and environments are designed and developed with consideration of economic factors and alternative technologies

For example:

- methods of product, system and technologies design, with consideration of markets, materials, sustainable factors, the environment and built-in obsolescence
- ways to appeal to consumers' desires when marketing products, services and environment in the local and regional community

WA9TDETS2

Critical and creative thinking Personal and social capability Ethical understanding

Design thinking skills

Project management

Manage projects, using suitable technologies, with an agile and collaborative approach. Use project management processes to consider time, risk, economic and sustainable factors

WA9TDEDTPM1

Numeracy Critical and creative thinking Personal and social capability

Investigating and defining

Ideate a problem and define the needs of an end user, through interviews and/or surveys

WA9TDEDTID1

Literacy Critical and creative thinking

Develop a design brief for a solution based on end user needs

WA9TDEDTID2

Critical and creative thinking

Investigate a range of technologies, resources and/or components to develop ideas and solutions, with consideration of social, ethical and other constraints

WA9TDEDTID3

Critical and creative thinking

Designing

Design alternative solutions considering available technologies, usability and aesthetics, using appropriate technical terms

WA9TDEDTD1

Critical and creative thinking

Producing and implementing

Select, implement and test a range of technologies, techniques and processes to produce designed solutions and/or prototypes

WA9TDEDTPI1

Literacy Critical and creative thinking Personal and social capability

Evaluating

Evaluate design processes and solutions against student-developed criteria

WA9TDEDTE1

Critical and creative thinking

Year 10

Year Level Description

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Design and Technologies builds on the earlier work students have experienced in investigating patterns, processes and phenomena, and explore forms of representation and use of a range of technologies. They understand that particular ways of managing, working and thinking have developed over time for particular reasons and may still be subject to critical review, revision and change. Learning experiences will enhance working collaboratively and enable students to draw on increasingly diverse and complex sources of information to facilitate comparing, contrasting, synthesising, questioning and critiquing information to achieve desired solutions.

In Year 10, students have opportunities to learn about technologies in society and ways people consider social, ethical, sustainable and security factors to adapt and improve design and production systems in at least one of the following Design and Technologies contexts: Engineering principles and systems, Food and fibre production, Food specialisations, and Materials and technologies specialisations. Students develop an understanding of ways products, services and environments are designed and developed. They consider specialised occupations and economic factors to identify market opportunities, innovate, create and develop entrepreneurial behaviours.

Students apply design thinking skills and use divergent techniques to generate design ideas for user experiences and a range of solutions. They manage and accommodate social, ethical, sustainable and consumer and/or producer considerations in the development of entrepreneurial and marketing strategies for specialised products, systems and environments. Students evaluate enterprise opportunities and impact of existing and new solutions. They competently identify and manage processes for effective project management with consideration of time, production processes, social, ethical, economic and sustainable factors, and legal responsibilities for optimum quality and performance to achieve designed solutions.

Achievement Standard

By the end of the year:

Students consider ways social, ethical, sustainable and security factors affect designed solutions, complexity of design, and production processes. In Engineering principles and systems, students identify the process of combining materials with force, motion and energy to design solutions. In Food and fibre production, students consider the role of technological innovations in ways food and fibre products are grown, processed and marketed in the design of sustainable products and systems. In Food specialisations, students select processing techniques for the preservation of food products, considering nutrition principles, consumer and/or producer values and ways properties influence design, preparation and development of specialised food products. In Materials and technologies specialisations, students consider the functional properties of materials and the application of specialised technologies and systems in the design and development of designed solutions.

In the Design and Technologies contexts, students ideate a problem and define the needs of the client/stakeholder to develop a design brief and determine the basis for a designed solution. They determine required resources and availability to develop designed solutions, considering associated constraints. Students consider specialised occupations and economic factors to identify market opportunities, innovate, create and develop entrepreneurial behaviours to design and develop products, services and environments for clients/stakeholders. Students provide relevant analysis to evaluate design processes and solutions against student-developed criteria. They manage projects, using suitable technologies, with an agile and collaborative approach. Students use project management processes/skills, consider time, production, social, ethical, economic and sustainable factors, and legal responsibilities.

Content Descriptions

Contexts

Engineering principles and systems

Effect of materials when combined with force, energy and/or motion in the design of ethical and sustainable engineered products and systems

Social, ethical, sustainable, consumer and producer considerations in the design and development of entrepreneurial and marketing strategies for an engineering enterprise, including management of risks, security measures and for optimum quality and performance to achieve designed solutions

For example:

- management of the development, production and evaluation of a programmable device using kits or from scratch for an 'engineered futures' exhibition
- engineered audio systems to develop individualised hearing aids, live music streaming online and reliable communications
- management of ethical factors and the use of automation and artificial intelligence (AI) in the design of engineered systems, considering employment opportunities in vehicle manufacturing, food processing and clothing construction
- automata, powered automatons
- stationary or mobile devices with multiple, simple actions
- perpetual motion machines

Literacy Critical and creative thinking Personal and social capability Ethical understanding

Food and fibre production

Role of technological innovations in ways food and fibre products are grown, processed and marketed, in the design of ethical and sustainable products and systems

Social, ethical, sustainable, consumer and producer considerations in the design and development of entrepreneurial and marketing strategies for a food- and/or fibre-based enterprise, including management of risks, security measures and regulatory responsibilities for optimum quality and performance to achieve designed solutions

For example:

• social
laboratory-developed protein sources, feed-lots for production, such as meat, eggs, wheat, cotton
robotics capable of picking, sorting, inspecting, packaging and transporting food and/or fibre products

• laboratory-developed protein sources, feed-lots for production, such as meat, eggs, wheat, cotton
• robotics capable of picking, sorting, inspecting, packaging and transporting food and/or fibre products
• ethical
fair employment in developing countries
greenwashing – when an organisation spends more time and money on marketing itself as environmentally friendly than on actually minimising its environmental impact
primary source of food or fibre composition and true labelling of food and fibre products
role of herbicide or chemicals and alternatives, to increase yield, as a component of an enterprising system

• fair employment in developing countries
• greenwashing – when an organisation spends more time and money on marketing itself as environmentally friendly than on actually minimising its environmental impact
• primary source of food or fibre composition and true labelling of food and fibre products
• role of herbicide or chemicals and alternatives, to increase yield, as a component of an enterprising system
• sustainable
quarantine to prevent pests, diseases and weeds affecting production
genetically modified food or fibre production
regenerative – restore and revitalise natural food and plant systems
United Nations Sustainable Development Goals, such as responsible consumption and production of food and fibre products (food waste, 'fast fashion')
the design, development, production of a food and/or fibre product for a 'living in the future' exhibition

• quarantine to prevent pests, diseases and weeds affecting production
• genetically modified food or fibre production
• regenerative – restore and revitalise natural food and plant systems
• United Nations Sustainable Development Goals, such as responsible consumption and production of food and fibre products (food waste, 'fast fashion')
• the design, development, production of a food and/or fibre product for a 'living in the future' exhibition

WA10TDECFF1

Critical and creative thinking Personal and social capability Ethical understanding

Food specialisations

Processing techniques and the preservation of food products, considering application of nutrition principles; ways sensory and physicalproperties of food influence the design, preparation and development of specialised food products

Social, ethical, sustainable, consumer and producer considerations in the design and development of entrepreneurial and marketing strategies for a specialised food enterprise, including management of risks, security measures and regulatory responsibilities for optimum quality and performance to achieve designed solutions

For example:

• identified properties that influence the design of sustainable food products and consider nature positive, carbon zero and regenerative elements, particularly for animal- or plant- sourced products
• product tampering – tamper-proof packaging; truth in labelling; product substitution in food preservation
• consumer and producer values, including ways food security, media role in food messaging, labelling and convenience influence food solutions
• management of the development, production and evaluation of a food product for a 'food futures' exhibition, considering social and ethical factors
• national supply chain security with a preference for local production and employment; 'buy local' or 'food miles' or marketing for new food enterprises
• integration of sustainable solutions could include
circular – products, such as packaging, and systems designed to be reused, repurposed, repaired, remanufactured or reassembled
regenerative – restore and revitalise natural food systems

• circular – products, such as packaging, and systems designed to be reused, repurposed, repaired, remanufactured or

reassembled
- regenerative – restore and revitalise natural food systems

WA10TDECFS1

Critical and creative thinking Personal and social capability Ethical understanding

Materials and technologies specialisations

Functional properties of materials, combined with components and application of specialised technologies and systems in the design and development of designed solutions

Social, ethical, sustainable, consumer and producer considerations in the design and development of entrepreneurial and marketing strategies for specialised materials-based enterprise, including management of risks, security measures and regulatory responsibilities for optimum quality and performance to achieve designed solutions

For example:

- management of the development, production and evaluation of a product designed to achieve a specific purpose for inclusion at the 'living in the future' exhibition
- integration of materials and specialised technologies in the design and production of jewellery for personal use or a specific market
- use of composite materials, such as plywood, reinforced plastics or a combination of two or more materials, fibres or fabrics and specialised technologies (tools and equipment) in the design and production of products for personal or community use or a specific market, such as swimwear or activewear, baby or toddler sleepwear or for day-today activity
- sustainable solutions

circular – products, systems and components designed to be reused, repurposed, repaired, remanufactured or reassembled
regenerative – restore and revitalise natural materials systems

- circular – products, systems and components designed to be reused, repurposed, repaired, remanufactured or reassembled
- regenerative – restore and revitalise natural materials systems

WA10TDECMT1

Literacy Critical and creative thinking Personal and social capability Ethical understanding

Technologies and society

People consider social, ethical, sustainable and security factors to improve design and production systems using specialised technologies to achieve designed solutions

For example:

- strategies for sustainable production systems are subject to competing demands (social, environmental, economic) and how these factors influence functionality and process layout
- entrepreneurial activity, enterprising behaviours, including problem-solving strategies and critical thinking, encourage the introduction of something new, useful and is usually innovative to capture a specific market or to achieve a purpose

WA10TDETS1

Personal and social capability Ethical understanding

Products, services and environments are designed and developed with consideration of specialised occupations, economic and environmental factors to identify market opportunities, innovate, create and develop entrepreneurial behaviours

For example:

- computer-controlled production systems and technologies, such as artificial intelligence (AI), design for disassembly and sustainable features of 3D printed products
- innovation of new fibres, composite materials, material combinations and ways technologies influence product design, such as synthetics replacing natural fibres, traditional materials and use of 'intelligent' textiles for personal clothes
- self-management skills, such as goal setting, time management, resource management, budgeting, collaboration and continuous evaluation, contribute to success

WA10TDETS2

Critical and creative thinking Personal and social capability Ethical understanding

Design thinking skills

Project management

Manage projects, using suitable technologies, with an agile and collaborative approach. Use project management processes to consider time, production processes, social, ethical, economic and sustainable factors, and legal responsibilities

WA10TDEDTPM1

Investigating and defining

Ideate a problem and define the needs of the client/stakeholder through anecdotal evidence and/or data gathering techniques

WA10TDEDTID1

Literacy Critical and creative thinking

Develop a design brief for a solution or to innovate an existing product, service or environment

WA10TDEDTID2

Critical and creative thinking

Investigate a range of technologies, resources and/or components to develop ideas and solutions, with consideration of social and ethical factors, legal responsibilities and competing constraints

WA10TDEDTID3

Critical and creative thinking

Designing

Design alternative solutions considering available technologies, functionality, accessibility, usability and aesthetics, using appropriate technical terms

WA10TDEDTD1

Critical and creative thinking

Producing and implementing

Select, justify, implement and test a range of technologies, techniques and processes to produce designed solutions and/or prototypes

WA10TDEDTPI1

Numeracy Critical and creative thinking Personal and social capability

Evaluating

Evaluate design processes and solutions against student developed criteria

WA10TDEDTE1

Critical and creative thinking